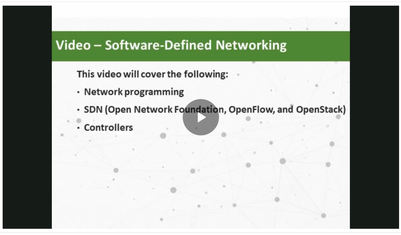
Software-Defined Networking

13.4.1

Video - Software-Defined Networking

Click Play to view a video on network programming, software-defined networking (SDN), and controllers.



13.4.2

Control Plane and Data Plane

The previous topic explained virtual network infrastructure. This topic will cover Software- Defined Networking (SDN). SDN was explained in the previous video. We will cover more details here.

A network device contains the following planes:

- **Control plane** – This is typically regarded as the brains of a device. It is used to make forwarding decisions. The control plane contains Layer 2 and Layer 3 route forwarding mechanisms, such as routing protocol neighbor tables and topology tables, IPv4 and IPv6 routing tables, STP, and the ARP table. Information sent to the control plane is processed by the CPU.
- **Data plane** – Also called the forwarding plane, this plane is typically the switch fabric connecting the various network ports on a device. The data plane of each device is used to forward traffic flows. Routers and switches use information from the control plane to forward incoming traffic out the appropriate egress interface. Information in the data plane is typically processed by a special data plane processor without the CPU getting involved.

Click each button for an illustration and explanation of the difference between the operation of localized control on a Layer 3 switch and a centralized controller in SDN

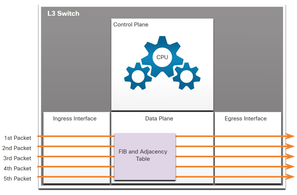
Layer 3 Switch and CEF

SDN and Central Controller

The figure illustrates how Cisco Express Forwarding (CEF) uses the control plane and data plane to process packets.

CEF is an advanced, Layer 3 IP switching technology that enables forwarding of packets to occur at the data plane without consulting the control plane. In CEF, the control plane's routing table pre-populates the CEF Forwarding Information Base (FIB) table in the data plane. The control plane's ARP table pre-populates the adjacency table. Packets are then forwarded directly by the data plane based on the information contained in the FIB and adjacency table, without needing to consult the information in the control plane.

Control Plane and Data Plane



Management Plane

Not shown in the figures is the management plane, which is responsible for managing a device through its connection to the network. Network administrators use applications such as Secure Shell (SSH), Trivial File Transfer Protocol (TFTP), Secure FTP, and Secure Hypertext Transfer Protocol (HTTPS) to access the management plane and configure a device. The management plane is how you have accessed and configured devices in your networking studies. In addition, protocols like Simple Network Management Protocol (SNMP), use the management plane.

13.4.3

Network Virtualization Technologies

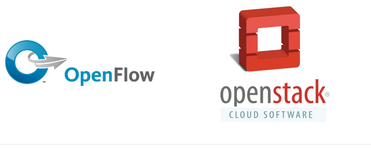
Over a decade ago, VMware developed a virtualizing technology that enabled a host OS to support one or more client OSs. Most virtualization technologies are now based on this technology. The transformation of dedicated servers to virtualized servers has been embraced and is rapidly being implemented in data center and enterprise networks.

Two major network architectures have been developed to support network virtualization:

- **Software-Defined Networking (SDN)** – A network architecture that virtualizes the network, offering a new approach to network administration and management that seeks to simplify and streamline the administration process.
- **Cisco Application Centric Infrastructure (ACI)** – A purpose-built hardware solution for integrating cloud computing and data center management.

Components of SDN may include the following:

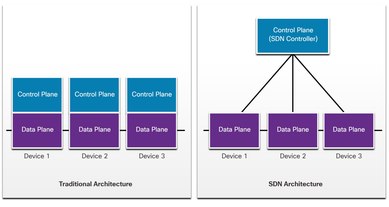
- **OpenFlow** – This approach was developed at Stanford University to manage traffic between routers, switches, wireless access points, and a controller. The OpenFlow protocol is a basic element in building SDN solutions. Search for OpenFlow and the Open Networking Foundation for more information.
- **OpenStack** – This approach is a virtualization and orchestration platform designed to build scalable cloud environments and provide an IaaS solution. OpenStack is often used with Cisco ACI. Orchestration in networking is the process of automating the provisioning of network components such as servers, storage, switches, routers, and applications. Search for OpenStack for more information.
- **Other components** – Other components include Interface to the Routing System (I2RS), Transparent Interconnection of Lotts of Links (TRILL), Cisco FabricPath (FP), and IEEE 802.1aq Shortest Path Bridging (SPB).



13.4.4

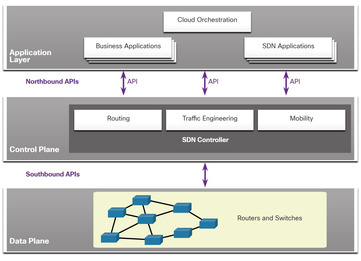
Traditional and SDN Architectures

In a traditional router or switch architecture, the control plane and data plane functions occur in the same device. Routing decisions and packet forwarding are the responsibility of the device operating system. In SDN, management of the control plane is moved to a centralized SDN controller. The figure compares traditional and SDN architectures.



The SDN controller is a logical entity that enables network administrators to manage and dictate how the data plane of switches and routers should handle network traffic. It orchestrates, mediates, and facilitates communication between applications and network elements.

The complete SDN framework is shown in the figure. Note the use of Application Programming Interfaces (APIs) within the SDN framework. An API is a set of standardized requests that define the proper way for an application to request services from another application. The SDN controller uses northbound APIs to communicate with the upstream applications. These APs help network administrators shape traffic and deploy services. The SDN controller also uses southbound APs to define the behavior of the data planes on downstream switches and routers. OpenFlow is the original and widely implemented southbound API.



13.4.5

Check Your Understanding - Software-Defined Networking

Check your understanding of SDN by choosing the BEST answer to the following questions.

1. Which characteristic best describes the control plane of a networking device? (Choose three.)

- [x] Information sent to this plane is processed by the CPU.
- [ ] This plane is also called the forwarding plane.
- [x] This plane makes forwarding decisions.
- [x] This plane contains router protocol neighbor and topology tables.
- [ ] This plane is responsible for switching frames out the interfaces.

2. Which characteristics best describe the data plane of a networking device? (Choose three.)

- [ ] This plane is the brains of the device.
- [x] This plane is used to forward traffic flows.
- [x] Information in this plane is typically processed by a special processor.
- [ ] This plane contains IPv4 and IPv6 routing tables.
- [x] This plane is typically made up of a switch fabric connecting the various ports on a device.

3. True of False. A network administrator accesses the control plane to configure a device.

- ( ) True
- (•) False

4. This network architecture offers a new approach to network administration and management that seeks to simplify and streamline the administration process.

- ( ) network virtualization
- (•) SDN
- ( ) Cisco ACI
- ( ) centralized controller

5. This is a logical entity that enables network administrators to manage and dictate how the data plane of switches and routers should handle network traffic.

- ( ) control plane
- ( ) data plane
- ( ) VM
- (•) SDN controller

6. This is a standardized request from the SDN controller to define the behavior of the data plane.

- ( ) North–South traffic
- ( ) Northbound APIs
- ( ) East-West traffic
- (•) Southbound APIs

Check

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